
MA 655: Fixed Income Securities

Course Code	MA 655
Course Name	Fixed Income Securities
L-T-P-C	2-1-0-3
Prerequisites	Knowledge of multivariable calculus, probability, statistics and stochastic process, differential equations and financial derivatives. Some knowledge of MATLAB/R/Spread sheets Packages.
Intended for	M.S./Ph.D.
Distribution	Elective
Approval	6th Senate (May 2014)

Course Contents

- **Introduction**

Basic concepts of fixed income instruments and their analysis, bond prices and yield curves, duration and convexity, empirical methods to find yield curve. [2 Lectures]

- **Pricing Theory and Models**

Asset pricing theory and related topics such as arbitrage, risk-neutral probability measures, martingale measures, hedging, intermediate dividends, complete and incomplete market etc., Factor models Single factor and multi factor diffusion model and their calibration, HW, BDT, and HJM models etc. [16 Lectures]

- **Credit derivatives and risk Management**

Market models, interest rate risk management, defaultable bonds and credit derivatives, mortgages and mortgage-backed Securities. [10 Lectures]

Text Books

1. Darrell Duffie, Kenneth J. Singleton, **Credit Risk**, Princeton University Press, 2003.
2. Sundaresan S., **Fixed Income Markets and Their Derivatives**, Academic Press, 3rd Ed., 2009.
3. Tuckman B. and Angel Serrat., **Fixed Income Securities**, John Wiley and Sons, 2011.

References

1. Brigo D. and F. Mercurio, **Interest Rate Models: Theory and Practice**, 2nd Edition, Springer Finance, 2006.
2. F. J. Fabozzi, **The Handbook of Fixed Income Securities**, McGraw-Hill, 2005.
3. Jarrow R., **Modeling Fixed Income Securities and Interest Rate Options**, McGraw-Hill, 1996.
4. Paul Wilmott, **Paul Wilmott on Quantitative Finance**, John Wiley & Sons, 2000.